

Optimal Bond-Equity Allocation Over a Lifetime

Mathematical Framework for Lifecycle Investing

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Outline

What is Lifecycle Investing?

Mathematical Framework

Human Capital Modeling

Optimal Allocation Derivation

Analytical Solution: Special Case

Extensions and Considerations

Applications and Takeaways

What is Lifecycle Investing?

Lifecycle Investing is the theory of optimal asset allocation over an individual's lifetime:

- ▶ **Core Principle:** Risk capacity changes with age, human capital, and time horizon
- ▶ **Key Insight:** Younger investors have more Human Capital (HC) (future earnings potential) which acts like a bond
- ▶ **Practical Application:** Age-based allocation rules and target-date funds

The fundamental trade-off:

- ▶ **Equities:** Higher expected returns but higher volatility
- ▶ **Bonds:** Lower returns but more stable, preservation of capital
- ▶ **Human Capital:** Bond-like asset that depletes over time

Classical heuristic: "100 minus age" rule suggests equity allocation
 $= 100 - \text{age}$

Mathematical Framework

Consider an investor with utility function U maximizing lifetime utility:

$$\max_{\{\alpha_t\}} \mathbb{E} \left[\sum_{t=0}^T \beta^t U(C_t) \right] \quad (1)$$

where:

- ▶ α_t = equity allocation at time t (bond allocation = $1 - \alpha_t$)
- ▶ C_t = consumption at time t
- ▶ β = time discount factor
- ▶ T = life expectancy (terminal age)
- ▶ R = retirement age (where $R < T$)

Wealth Dynamics

Total wealth consists of financial and human capital:

$$W_t = F_t + H_t \quad (2)$$

where F_t is financial capital and H_t is human capital (present value of future earnings)

Human Capital Modeling

Human capital represents the present value of future labor income:

$$H_t = \sum_{s=t+1}^R \frac{\mathbb{E}[Y_s]}{(1 + r_f)^{s-t}} \quad (3)$$

where:

- ▶ Y_s = labor income at time s
- ▶ R = retirement age
- ▶ r_f = risk-free rate

Properties of Human Capital

- ▶ Decreases with age (fewer working years remaining)
- ▶ Acts as an implicit bond holding (stable, regular cash flows)
- ▶ Cannot be traded but affects optimal portfolio allocation
- ▶ Risk characteristics depend on occupation (stable vs volatile income)

Young investors with large H_t can afford more equity exposure in F_t

Optimal Allocation Derivation - Part 1

Assume Constant Relative Risk Aversion (CRRA) utility: $U(C) = \frac{C^{1-\gamma}}{1-\gamma}$
where γ is risk aversion.

Portfolio Return Dynamics:

$$R_p = \alpha_t R_e + (1 - \alpha_t) R_b \quad (4)$$

Wealth Evolution:

$$F_{t+1} = (F_t - C_t)(1 + R_p) + Y_{t+1} \quad (5)$$

First Order Condition (Merton's Solution): For log-normal returns and CRRA utility, the optimal equity share is:

$$\alpha_t^* = \frac{\mu_e - r_f}{\gamma \sigma_e^2} \cdot \frac{W_t}{F_t} \quad (6)$$

where $\mu_e - r_f$ is the equity risk premium and σ_e^2 is equity variance

Optimal Allocation Derivation - Part 2

Key Insight: The ratio $\frac{W_t}{F_t} = 1 + \frac{H_t}{F_t}$ amplifies equity allocation when human capital is large.

Simplified Age-Based Formula: Assuming exponentially declining human capital and growing financial capital:

$$\alpha_t^* = \frac{\mu_e - r_f}{\gamma \sigma_e^2} \left(1 + \frac{H_0 e^{-\lambda t}}{F_0 e^{gt}} \right) \quad (7)$$

where:

- ▶ λ = human capital decay rate
- ▶ g = financial capital growth rate
- ▶ H_0, F_0 = initial human and financial capital

Result

Optimal equity allocation decreases with age as $\frac{H_t}{F_t} \rightarrow 0$, creating a natural "glide path"

Analytical Solution: Special Case

For tractability, consider the case with:

- ▶ Log utility ($\gamma = 1$)
- ▶ No intermediate consumption
- ▶ Deterministic human capital

The optimal allocation becomes:

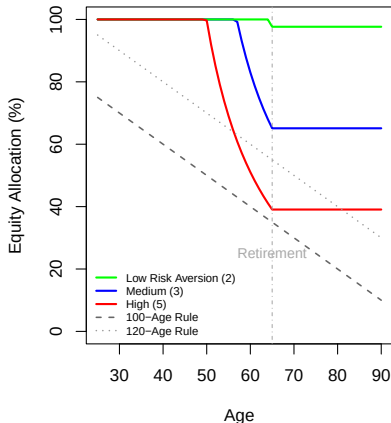
$$\alpha_t^* = \frac{\mu_e - r_f}{\sigma_e^2} \cdot \frac{W_t}{F_t} = \frac{\mu_e - r_f}{\sigma_e^2} \left(1 + \frac{H_t}{F_t} \right) \quad (8)$$

Boundary Conditions:

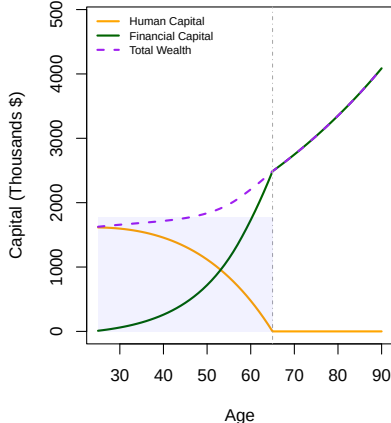
- ▶ At $t = 0$ (young): Large $H_0/F_0 \Rightarrow$ high α_0^* (potentially $> 100\%$)
- ▶ At $t = R$ (retirement): $H_R = 0 \Rightarrow \alpha_R^* = \frac{\mu_e - r_f}{\sigma_e^2}$ (constant)
- ▶ Allocation decreases monotonically with age

Visualization of Lifecycle Allocation

Optimal Equity Allocation by Risk Avers



Human vs Financial Capital Over Lifetime



- ▶ Left: Optimal allocation decreases with age as human capital depletes
- ▶ Right: Human capital dominates when young, financial capital dominates when old
- ▶ Model suggests higher equity allocation than simple heuristics for young investors

Extensions and Considerations

Model Extensions:

- ▶ **Stochastic Income:** Labor income volatility affects human capital risk
- ▶ **Housing:** Real estate as additional asset class with consumption value
- ▶ **Correlation:** Human capital may correlate with equity returns (industry-specific)
- ▶ **Flexibility:** Option value of delaying retirement

Practical Constraints:

- ▶ **Leverage Constraints:** Cannot borrow against human capital
- ▶ **Liquidity Needs:** Emergency funds require bond allocation
- ▶ **Behavioral Factors:** Risk tolerance may differ from optimal risk capacity
- ▶ **Tax Considerations:** Asset location across taxable/tax-deferred accounts

Implementation

Target-date funds and robo-advisors implement simplified versions of lifecycle models

Applications and Takeaways

Key Applications

- ▶ **Retirement Planning:** Design of target-date funds and 401(k) defaults
- ▶ **Robo-Advisors:** Automated age-based rebalancing strategies
- ▶ **Financial Advice:** Framework for personalized asset allocation
- ▶ **Policy Design:** Optimal default options for pension systems

Takeaways

- ▶ Human capital acts as an implicit bond holding, justifying higher equity allocation when young
- ▶ Optimal allocation follows: $\alpha_t^* = \frac{\mu_e - r_f}{\gamma \sigma_e^2} \left(1 + \frac{H_t}{F_t} \right)$
- ▶ Traditional rules like "100-age" may be too conservative for young investors
- ▶ Risk capacity $\neq$ risk tolerance: psychological factors matter
- ▶ Lifecycle investing provides a rigorous framework beyond simple heuristics